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Arithmetic Sequences and Series

Add the same d every time. Two formulas do everything: the n th term and the sum. Two facts give two equations in a and d .

THE TOPIC ON ONE MAP

Arithmetic Sequences and Series

01 — The n th term

$$a + (n - 1)d$$

two terms $\rightarrow a, d$

algebraic terms

02 — The sum

$$\frac{n}{2}(2a + (n - 1)d)$$

$$\frac{n}{2}(a + l)$$

quadratic in n

03 — In context

year n total after N years

rows of seats

THE TWO FORMULAS

$$u_n = a + (n - 1)d \quad S_n = \frac{n}{2}\{2a + (n - 1)d\} = \frac{n}{2}(a + l)$$

a first term d common difference l last term n number of terms (a whole number)

01 The n th Term: $u_n = a + (n - 1)d$

Any term is the first term plus $(n - 1)$ steps of d . Two terms given means two equations: subtract them to find d . Three consecutive terms in algebra: the middle one is the average of its neighbours.

WHEN YOU SEE → YOU DO

“the 4th and 8th terms are 20 and 64”

→ $a + 3d = 20, a + 7d = 64$: subtract

“ $(p - 9), -7, (9 - 3p)$ are the first three terms”

→ $2(\text{middle}) = \text{first} + \text{third}$

“the number of terms in $2 + 5 + 8 + \dots + 254$ ”

→ $254 = 2 + (n - 1)3$

“find the 18th term”

→ $a + 17d$

WORKED MODEL · JANUARY 2025

$2 + 5 + 8 + 11 + \dots + 254$. How many terms?

1 $254 = 2 + (n - 1) \times 3$

last term

ANSWER $n - 1 = 84 \Rightarrow n = 85$

IDEA 01

Practise: The n th Term: $u_n = a + (n - 1)d$

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Two terms given

P2 · TOPIC 5 3 marks

The 4th term of an arithmetic sequence is 17 and the 9th term is 42. Find a and d .

Subtract the two equations: $5d = 25$.

Q2 DR ESLAM Algebraic terms

P2 · TOPIC 5 3 marks

x , $2x + 1$, $5x - 2$ are consecutive terms of an arithmetic sequence. Find x .

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P2 · TOPIC 5

2 marks

1 Write down the next three terms in these arithmetic sequences

(i) 30, 18, 6, ...

(ii) $\frac{1}{4}, \frac{5}{12}, \frac{7}{12}, \frac{3}{4}, \dots$

(2 marks)**Q4****PRACTICE**

Practice · Medium

P2 · TOPIC 5

4 marks

3 An arithmetic sequence has first term r^2 and common difference $3r$, where $r > 0$. The fifth term of the sequence is 85.

Find:

(i) the value of r

(ii) the ninth term in the sequence.

(4 marks)

Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 5 4 marks

2 The first three terms in an arithmetic sequence are $2, 3q^2, q, \dots$

Given that the first three terms in the sequence are all positive, find the fortieth term in the sequence.

(4 marks)

PAST PAPERS

Q6 **PAST PAPER** Pearson IAL WMA12 · January 2025 · Q1**P2 · TOPIC 5** **4 marks**

1. The arithmetic series S is given by

$$S = 2 + 5 + 8 + 11 + \dots + 254$$

Find

- (a) the number of terms in the series, (2)
- (b) the sum of the series. (2)

(Total for Question 1 is 4 marks)

Q7**PAST PAPER**

Pearson IAL WMA12 · October 2025 · Q1

P2 · TOPIC 5 4 marks

1. An arithmetic series starts

$$78 + 75 + 72 + \dots$$

Find

- (a) the 18th term in the series, (2)

- (b) the sum of the first 100 terms of the series. (2)

(Total for Question 1 is 4 marks)

02 The Sum: $S_n = \frac{n}{2}\{2a + (n - 1)d\}$

Use $\frac{n}{2}(a + l)$ when you know the last term, the long form otherwise. A sum that is given as a number, with n unknown, becomes a quadratic in n : keep the positive whole-number root.

WHEN YOU SEE → YOU DO

“the sum of the first ten terms is -80”

→ $5(2a + 9d) = -80$

“the sixth term is 2” and a sum

→ two equations in a, d

“find the smallest n for which $S_n > 8000$ ”

→ solve $S_n = 8000$, then round **up**

“show that $S_n = \frac{n}{2}\{2a + (n - 1)d\}$ ”

→ write the series forwards and backwards, add

“900 + 892 + 884 + ... + 500”

→ n from the last term, then $\frac{n}{2}(a + l)$

Write the facts

Two equations

Solve for a and d

Answer the question

WORKED MODEL · JUNE 2024

$u_6 = 2$ and $S_{10} = -80$. Find a and d .

1 $a + 5d = 2$ and $5(2a + 9d) = -80 \Rightarrow 2a + 9d = -16$

two equations

ANSWER $d = 20, a = -98$

× **TRAP** Giving $n = 34.6$ as the answer. n counts terms: round up for “exceeds”, and check it.

IDEA 02

Practise: The Sum: $S_n = \frac{n}{2}\{2a + (n - 1)d\}$

My questions first, then selected practice, then real past papers.

Q8 DR ESLAM A plain sum

P2 · TOPIC 5 2 marks

Find the sum of the first 20 terms of $5 + 8 + 11 + \dots$

Q9 DR ESLAM Smallest n

P2 · TOPIC 5 4 marks

Find the smallest n for which $3 + 7 + 11 + \dots$ (to n terms) exceeds 1000.

SELECTED PRACTICE

Q10**PRACTICE**

Practice · Easy

P2 · TOPIC 5 2 marks

- 2 Find the sum of the first four terms in the sequence defined by $u_n = 2n + 3$.
Justify why this sequence is an arithmetic sequence.

(2 marks)

Q11**PRACTICE**

Practice · Medium

P2 · TOPIC 5 9 marks

8 (a) Calculate the sum of all the odd numbers between 0 and 150,

$$1 + 3 + 5 + \dots + 149$$

(3 marks)

(b) An arithmetic series is defined by

$$k + 2k + 3k + \dots + 360$$

where k is an integer and a positive factor of 360.

(i) In terms of k , find an expression for the number of terms in this series.

(ii) Show that the sum of this series is $180 + \frac{64800}{k}$.

(4 marks)

(c) In terms of q , find the 100th term of the arithmetic sequence defined by

$$(3q - 7), (5q - 4), (7q - 1), \dots$$

Give your answer in simplest form.

(2 marks)

Q12**PRACTICE**

Practice · Very Hard

P2 · TOPIC 5 4 marks

- 5 The sum of the first 24 terms in an arithmetic series is nine times the sum of the first two terms in the series.

Find the sum of the first 90 terms in the series.

(4 marks)

PAST PAPERS

Q13 PAST PAPER Pearson IAL WMA12 · June 2024 · Q2

P2 · TOPIC 5 7 marks

2. **In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

In an arithmetic series,

- the sixth term is 2
- the sum of the first ten terms is -80

For this series,

- (a) find the value of the first term and the value of the common difference. (4)

- (b) Hence find the smallest value of n for which

$$S_n > 8000 \quad (3)$$

(Total for Question 2 is 7 marks)

Q14**PAST PAPER**

Pearson IAL WMA12 · June (R paper) 2024 · Q8

P2 · TOPIC 5 13 marks

8. (i) (a) In an **arithmetic** series the first term is a and the common difference is d .

Show that

$$S_n = \frac{n}{2} \{2a + (n-1)d\} \quad (3)$$

- (b) Hence find

$$900 + 892 + 884 + \dots + 500 \quad (3)$$

- (ii) Given that the first three terms of a **geometric** series are

$$k+4 \quad k-2 \quad 11-k$$

where k is a constant,

- (a) show that

$$2k^2 - 11k - 40 = 0 \quad (3)$$

Given also that this series is convergent,

- (b) find the value of S_∞ (4)

(Total for Question 8 is 13 marks)

03 Arithmetic Series in Context

Year 1, row 1, month 1 is the first term. “Each year ...less than the previous” is $d < 0$. “Mass extracted in year 14” is a term; “total extracted” is a sum.

WHEN YOU SEE → YOU DO

“480 tonnes in year 1, 465 in year 2 ...”

→ $a = 480, d = -15$

“the mass extracted in year 14”

→ u_{14}

“after a total of 7770 tonnes ...at the end of year N ”

→ $S_N = 7770$: quadratic in N

“each row has 4 more seats”

→ $d = 4$

“hence state the value of N ”

→ the root that makes sense

WORKED MODEL · OCTOBER 2021

480, 465, 450, ... tonnes. Total 7770 tonnes after N years. Show $N^2 - 65N + 1036 = 0$.

$$1 \quad \frac{N}{2}\{960 - 15(N - 1)\} = 7770$$

sum

$$2 \quad N(975 - 15N) = 15540 \Rightarrow 15N^2 - 975N + 15540 = 0$$

expand

$$\text{ANSWER} \quad N^2 - 65N + 1036 = 0 \Rightarrow N = 28 \text{ (37 gives negative years)}$$

IDEA 03

Practise: Arithmetic Series in Context

My questions first, then real past papers.

Q15 **DR ESLAM** Savings plan

P2 · TOPIC 5 3 marks

Sam saves £50 in month 1 and £10 more each month than the month before. How much has Sam saved after 2 years?

PAST PAPERS

Q16 PAST PAPER Pearson IAL WMA12 · October 2021 · Q5

P2 · TOPIC 5 6 marks

5. A company that owned a silver mine

- extracted 480 tonnes of silver from the mine in year 1
- extracted 465 tonnes of silver from the mine in year 2
- extracted 450 tonnes of silver from the mine in year 3

and so on, forming an arithmetic sequence.

(a) Find the mass of silver extracted in year 14

(2)

After a total of 7770 tonnes of silver was extracted, the company stopped mining.

Given that this occurred at the end of year N ,

(b) show that

$$N^2 - 65N + 1036 = 0$$

(3)

(c) Hence, state the value of N .

(1)

(Total 6 marks)

Q17 **PAST PAPER** Pearson IAL WMA12 · October 2023 · Q8**P2 · TOPIC 5** **7 marks**

8. In a large theatre there are n rows of seats, where n is a constant.

The number of seats in the first row is a , where a is a constant.

In each subsequent row there are 4 more seats than in the previous row so that

- in the 2nd row there are $(a + 4)$ seats
- in the 3rd row there are $(a + 8)$ seats
- the number of seats in each row form an **arithmetic** sequence

Given that the **total** number of seats in the first 10 rows is 360

(a) find the value of a . (2)

Given also that the total number of seats in the n rows is 2146

(b) show that (2)

$$n^2 + 8n - 1073 = 0$$

(c) Hence (3)

- (i) state the number of rows of seats in the theatre,
- (ii) find the maximum number of seats in any one row.

(Total for Question 8 is 7 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|------------------------|---|
| 01 | <i>n</i>th term | $a + (n - 1)d$; two terms, subtract; middle = average. |
| 02 | Sum | $\frac{n}{2}\{2a + (n - 1)d\}$ or $\frac{n}{2}(a + l)$; n is a whole number. |
| 03 | Context | Year 1 = first term; term vs total. |

BOTTOM LINE Name a and d · term or sum? · two facts → two equations
· n must be a whole number.

Answers

Arithmetic Sequences and Series · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · THE n TH TERM: $u_n = a + (n - 1)d$

- Q1** Dr Eslam · Two terms given $d = 5, a = 2$
- Q2** Dr Eslam · Algebraic terms $2(2x + 1) = x + (5x - 2) \Rightarrow x = 2$
- Q3** Practice · Easy (i) $-6, -18, -30$ (ii) $d = \frac{1}{6}; \frac{11}{12}, \frac{13}{12}, \frac{5}{4}$
- Q4** Practice · Medium (i) $r^2 + 12r = 85 \Rightarrow r = 5$ (ii) $u_9 = 25 + 8(15) = 145$
- Q5** Practice · Very Hard $6q^2 = 2 + q \Rightarrow q = \frac{2}{3}$ (positive); $d = -\frac{2}{3}; u_{40} = 2 + 39(-\frac{2}{3}) = -24$
- Q6** Pearson IAL WMA12 · January 2025 · Q1 (a) 85. (b) 10880
- Q7** Pearson IAL WMA12 · October 2025 · Q1 (a) 27. (b) -7050

IDEA 02 · THE SUM: $S_n = \frac{n}{2}\{2a + (n - 1)d\}$

- Q8** Dr Eslam · A plain sum $10(10 + 57) = 670$
- Q9** Dr Eslam · Smallest n $S_n = 2n^2 + n > 1000$: $n = 22$ gives 990, $n = 23$ gives 1081: $n = 23$
- Q10** Practice · Easy $5 + 7 + 9 + 11 = 32$; the difference $u_{n+1} - u_n = 2$ is constant
- Q11** Practice · Medium (a) 75 terms: 5625 (b)(i) $\frac{360}{k}$ (ii) $\frac{360}{2k}(k + 360) = 180 + \frac{64800}{k}$ (c) $d = 2q + 3$: $201q + 290$
- Q12** Practice · Very Hard $12(2a + 23d) = 9(2a + d) \Rightarrow a = -44.5d$; $S_{90} = 45(2a + 89d) = 0$
- Q13** Pearson IAL WMA12 · June 2024 · Q2 (a) First term = -98, common difference = 20. (b) $n = 35$
- Q14** Pearson IAL WMA12 · June (R paper) 2024 · Q8 (i)(a) Proven. (i)(b) 35700. (ii)(a) Proven. (ii)(b) $S_\infty = 24$

IDEA 03 · ARITHMETIC SERIES IN CONTEXT

- Q15** Dr Eslam · Savings plan $a = 50, d = 10, n = 24$: $12(100 + 230) = £3960$
- Q16** Pearson IAL WMA12 · October 2021 · Q5 (a) 285 tonnes. (b) Proven: $N^2 - 65N + 1036 = 0$. (c) $N = 28$
- Q17** Pearson IAL WMA12 · October 2023 · Q8 (a) $a = 18$. (b) Proven. (c)(i) 29 rows. (c)(ii) 130 seats